

Finding a Convex Hull subject to Satisfaction of Constraints



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Yes. If by “**convex hull with constraint satisfaction**” you mean:

Find the smallest convex hull / convex polygon containing a set of candidate points while satisfying additional constraints

there are several established formulations. The terminology varies depending on what the constraints are.

1. Convex-hull constraint as an optimization problem

A particularly clean formulation is to require the solution x to lie inside the convex hull of given points $b_1, \dots, b_m$:

$$x \in \text{conv}\{b_1, \dots, b_m\}.$$

This can be represented using non-negative barycentric weights:

$$x = \sum_{i=1}^m w_i b_i,$$

with

$$w_i \geq 0, \sum_{i=1}^m w_i = 1.$$

Thus a convex-hull membership constraint becomes a set of **linear equality and inequality constraints**. This formulation is used directly in convex optimization. [\(GitHub\)](#)

2. Convex hull of constraint sets

If your constraints define several feasible regions

$$F_1, F_2, \dots, F_k,$$

you can explicitly construct

$$\text{conv}(F_1 \cup F_2 \cup \dots \cup F_k).$$

For example, YALMIP has a `hull()` construction that formulates the convex hull of constraint-defined sets using auxiliary/lifting variables. [\(YALMIP\)](#)

This is particularly relevant to **constraint satisfaction / mixed-integer optimization**, because a logical disjunction such as

$$x \in F_1 \vee x \in F_2$$

can be replaced by a convex-hull formulation.

3. Constraints on the hull itself

A more interesting problem is:

$$\text{find } H \text{ such that } P \subseteq H, H \text{ is convex, } H \text{ satisfies constraints } C_1, C_2, \dots, C_k.$$

Examples of constraints could be:

- maximum area
- maximum perimeter
- fixed number of vertices
- specified bounding box
- forbidden regions
- required regions
- minimum/maximum distance from objects
- prescribed edge orientations
- minimum internal angle
- land-use restrictions
- connectivity
- exclusion of particular points/areas.

There is research specifically on constrained polygonization and generalized convex-hull problems. For example, constrained α -polygon formulations use assignment variables and constraints to ensure that points are assigned to polygon edges, that assignments don't conflict, that the polygon is simple, and that angle constraints are satisfied. When $\alpha=0$, the formulation reduces to the ordinary convex hull. [\(MDPI\)](#)

4. Vertex → inequality representation

If you already have candidate hull vertices V , another useful approach is to convert them into inequalities:

$$H=\{x:Ax\leq b\}.H=\{x:Ax\leq b\}.$$

The resulting inequalities define the tight convex polytope containing the vertices. MATLAB's **VERT2CON**, for example, performs exactly this conversion and removes redundant constraints. ([MathWorks](#))□

This gives a natural framework for adding additional constraints:

$$Ax\leq b,Cx\leq d\boxed{Ax\leq b,\,\,\,\,\,\,Cx\leq d}$$

where $Ax\leq b$ describes the convex hull and $Cx\leq d$ represents your external constraints.

If you're looking for an algorithm rather than just the mathematical formulation

I think the problem you may be getting at is more specific:

Find a convex hull satisfying a collection of spatial constraints

For example, given thousands of GIS points/polygons:

$$P=\{p_1,\dots,p_n\},P=\{p_1,\ldots,p_n\},$$

find a convex polygon H such that

$$P_{\text{required}}\subseteq H, P_{\text{forbidden}}\cap H=\emptyset, P_{\text{forbidden}}\cap H=\text{varnothing}, A_{\min}\leq \text{Area}(H)\leq A_{\max}, A_{\min}\leq \text{Area}(H)\leq A_{\max}, \text{perimeter}(H)\leq L, \text{perimeter}(H)\leq L, H\subseteq B, H\subseteq B,$$

and perhaps

$$\text{distance}(H,F_i)\geq d_i.\text{distance}(H,F_i)\geq d_i.$$

That becomes a **constrained convex-hull optimization / computational geometry problem**, and the appropriate method could be LP, QP, MILP, or a specialized geometric algorithm depending on the constraints.

If you tell me the **specific constraints you want imposed on the convex hull**, I can search specifically for the closest existing algorithm/paper and formulate the problem mathematically.